

BEYZA AKGÜN

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

SUMMARY

Computer Engineering graduate focused on backend development with Java, Spring Boot, Python, and FastAPI. Hands-on project experience developing REST APIs, implementing JWT authentication, and building applications with PostgreSQL and Redis. Applied JUnit and Mockito for backend testing and built real-time multiplayer features using WebSockets. Brings enterprise IT troubleshooting experience from KoçSistem and a strong foundation in object-oriented programming and relational databases.

EXPERIENCE

Software Unit Management Intern — Prodea Bilgi ve Danışmanlık | Jan 2025 – Feb 2025

- Contributed to bug fixes and code improvements in an existing enterprise codebase.
- Supported technical documentation and task tracking within the development team.

Microsoft Operating Systems Project Assistant — KoçSistem | Mar 2024 – Sep 2024

- Resolved 50+ incidents across Active Directory, DNS, DHCP, Group Policy, and Microsoft 365 through systematic troubleshooting and PowerShell.
- Supported CrowdStrike incident response and service restoration in coordination with technical teams.
- Documented incidents, tracked resolutions, and escalated complex issues to relevant teams.

PROJECTS

Country Guessing Game | *Python, FastAPI, PostgreSQL, SQLAlchemy, Redis, Alembic, JWT, WebSockets, React, TypeScript*

- Built a geography game with 8 modes, JWT authentication, global leaderboards and 7 rank tiers.
- Enabled ranked 1v1 matches via WebSockets with reconnect support and Redis matchmaking.
- Developed REST APIs with async SQLAlchemy/asyncpg and Alembic migrations; deployed the backend on Fly.io and frontend on Vercel.

Airline Operations Dashboard | *Java, Spring Boot, PostgreSQL, Angular, TypeScript, RxJS, Playwright, GitHub Actions*

- Developed 7 REST API operations with PostgreSQL persistence, DTO validation, unique flight number checks and centralized exception handling.
- Built an Angular dashboard with flight search, filtering, CRUD operations and 4 summary cards using Reactive Forms, RxJS and Signals.
- Integrated 24 tests (15 backend, 7 frontend, 2 E2E) into a 3-job CI pipeline, using Testcontainers for PostgreSQL and Playwright for core flight workflows.

Bubble Sheet Scanner Using Deep Learning | *Python, PyTorch, Hugging Face Transformers, OpenCV, Semantic Segmentation, Image Captioning*

- Developed a template-free OMR pipeline combining semantic segmentation and image captioning to extract exam answers.
- Evaluated 3 segmentation architectures on 2,814 exam papers and 5,454 cropped regions, achieving a 0.9996 Dice score with SegFormer.
- Built an EfficientNet-B0/Transformer model achieving 71.85% token accuracy and 52.40% exact-match accuracy versus 14.90% for the best evaluated MiniCPM-V configuration.

EDUCATION

Bachelor of Science in Computer Engineering | Istanbul Bilgi University | 2021 – 2025

GPA: 3.21/4.00 | 50% Merit Scholarship | Dean's Honor List (every semester)

Best Senior Project Award — Bubble Sheet Scanner Using Deep Learning

SKILLS

Programming Languages: Java, Python, SQL, TypeScript, JavaScript

Backend: Spring Boot, FastAPI, Spring Data JPA, Hibernate, SQLAlchemy, REST APIs, WebSockets, JWT

Databases & Caching: PostgreSQL, Redis

Testing & CI/CD: JUnit, Mockito, Vitest, Pytest, Playwright, Testcontainers, GitHub Actions

Tools: Git, Docker, Linux, Maven, Postman

Frontend: Angular, React, RxJS, HTML5, CSS3

AI/ML & Data: PyTorch, Scikit-learn, Transformers, OpenCV, Pandas, NumPy, Matplotlib, Seaborn

Languages: Turkish (Native), English (C1)